



SENTINEL NEXUS OPERATOR MANUAL

Absolute guidance for the Nexus intelligence fabric

Nexus is the operating intelligence layer for SentinelOps. It connects incidents, services, databases, dependency topology, business flows, light agents, SOPs, AI interpretation, safe diagnostics, guarded restart, rollover, and operator verdicts into one transparent command experience.

START
Incidents

DEFINE
Services

MODEL
Flows

ACT
Diagnostics

TEACH
Verdicts

What Sentinel Nexus is and how to enter it

Nexus is not a normal monitoring page. Network Sentinel proves reachability. Nexus decides what the reachability means by combining service contracts, runtime evidence, dependencies, databases, business flows, SOPs, and operator outcomes.

INCIDENTS

Live work

SERVICES

Truth units

DATABASES

Root causes

ACTIONS

Safe lanes

OUTCOME

Learning

FIRST THING TO READ

Nexus command map

Start by reading fabric pulse, sync health, mapped service count, database contracts, active incidents, diagnostics-ready services, restart-ready services, and current workspace tab.

- Use Incidents when production is actively noisy.
- Use Services when Nexus seems to misunderstand a service.
- Use Databases when the evidence mentions sessions, pools, locks, ORA, TNS, SQLSTATE, JDBC, storage, or replication.
- Use Agents when runtime evidence is missing, stale, or mapped to the wrong service.
- Use SOPs when operators need governed procedure guidance.

Rule: If the catalog is wrong, Nexus will reason from the wrong world. Fix catalog truth before blaming the engine.

NEXUS WORKSPACE**Command map****Incidents****Services****Agents****Databases****Rollover****SOPs**

Mapped services

28 / 31

Diagnostics lane

12 ready

Business flows

9 active

Database contracts

7 modeled**Fabric pulse**

Sync success

Linked consoles

Network, telemetry, evidence

Operator loop

Observe, correlate, decide, act, learn

Investigate from risk, not noise

The Incidents workspace is where Nexus turns signals into an operational story. Open the highest-risk active incident first, then move through overview, topology, evidence, actions, procedure, and outcome.

**INCIDENT
DETAIL**

**Risk and root
candidates**

CRITICAL **Risk 91** **IDC access flow**

Status OPEN

Affected
**IDC
Gateway**

Suspected
root
ARX Auth

Confidence
0.82

01 ARX Auth Upstream, auth failures, strong fit

02 IDC Gateway Symptomatic, runtime healthy

03 Core DB Low evidence fit

CHECKLIST

How to triage the incident

1. Read title, risk level, risk score, business impact score, status, affected services, and suspected root.
2. Confirm which business flows and clusters are attached. This tells you who is affected, not just what is noisy.
3. Open Overview first for the story, then Topology for direction, Evidence for proof, Actions for safe moves, Procedure for SOP guidance, and Outcome for verdict.
4. Use status honestly: OPEN for active work, MONITORING for watch state, AWAITING_VERDICT after action, RESOLVED only after outcome is recorded.

Example: An ARX problem can explain IDC user access without explaining IDC transaction posting if the ARX dependency is flow-scoped to access only.

Read before action

Risk, affected services, suspected root, confidence, blast radius, evidence timeline, signatures, recent changes, and recommendations must agree enough to justify the next move.

Do not chase alerts blindly

The newest alert is not always the root. Nexus ranks by graph, business flow, evidence diversity, upstream explanation, and business impact.

Use iNterpret for speed, evidence for authority

The Nexus AI brief translates evidence into operator language. It is valuable, but deterministic correlation and cited evidence remain the control plane. The operator must separate known facts, inference, uncertainty, and unsafe action.

EVIDENCE LANE

Timeline before action

10:01external_network timeout

10:03agent heartbeat healthy

10:05ORA-00020 sessions high

10:07pool used/max critical

10:09slow query count rising

Source
Network Sentinel

Source
Light agent

Source
DB probe

NEXUS AI BRIEF

Explain, cite, constrain

KnownExternal checks fail on gateway.

KnownRuntime heartbeat is healthy.

InferredDatabase pool pressure explains timeouts.

UnsafeRestart blocked until diagnostics.

Ask precise questions: procedure, validation, rollback, escalation, or risk. Avoid vague "what is wrong?" questions when the incident already has structured evidence.

EVIDENCE METADATA

Every signal must carry meaning

- service_id, service.name, environment, instance_id when known.
- timestamp, severity, source, vantage_point, observation_layer.
- failure_domain_hint when known: network, runtime, database, dependency, auth, integration, storage, change.
- Log signature fields: error class, exception name, timeout type, OOM flag, DB code, first seen, last seen, count, representative snippets.

DATABASE EVIDENCE

Preserve the codes

- Oracle: ORA, TNS, sessions, locks, blocking sessions, listener, tablespace, replication, slow queries.
- PostgreSQL: SQLSTATE, connection errors, lock waits, replication lag, storage pressure, slow query count.
- JDBC and pool evidence: pool used/max, max pool size, connection timeout, statement timeout, transaction state.

Diagnostics first, task handoff when humans must own it, restart only through the gate

Nexus action lanes are built for safety. They give operators a clear path without turning the page into arbitrary remote control.

ACTION DECISION

Choose the truthful move

- **Request diagnostics** when the target is plausible but more safe evidence is needed.
- **Create response task** when a person or team needs accountable follow-up.
- **Approve restart** only when Nexus policy gates, evidence, and service certification all line up.
- **Reject restart** when the recommendation is unsafe, premature, stateful, low confidence, or dependency-driven.

Never restart db, cache, queue, auth, infra, shared databases, core banking failover paths, or config-change paths from Nexus.

ACTIONS TAB

Evidence-gated control

Request diagnostics

Create task

Restart locked

Required
diagnostics_ready

Required
stateless

Required
cooldown
clear

Restart policy

allow_restart true

Allowed type

app or gateway only

Approval

human challenge complete

Diagnostics checklist

- Service is diagnostics_ready or restart_ready.
- Agent or diagnostics URL is configured.
- Output is bounded and read-only.
- Result explains host, logs, sockets, memory, disk, dependencies, and DB probes where safe.

Restart checklist

- Service is restart_ready and stateless.
- Policy allows restart for this service type.
- No maintenance block or cooldown is active.
- Confidence is high enough and evidence supports restart as safer than waiting.

Build the service contract before you expect intelligence

A service is the smallest unit Nexus can reason about. The service contract is where identity, ownership, evidence, policy, and certification become explicit.

SERVICES WORKSPACE

Service contract

service_id idc-gateway-prod

service_type gateway

environment production

owner_team Digital Core

criticality critical

network_service_id net-idc-gateway

agent_id agent-prod-02

certification diagnostics_ready

REQUIRED FIELDS

What to configure

- Stable ID, clear name, environment, service type, owner team, criticality, description, tags.
- Cluster IDs and legacy cluster label where applicable.
- Endpoint config: healthcheck, metrics, logs, traces, diagnostics, restart, extraction, formatting, shipping, dashboard.
- Observation config: Network Sentinel mapping, agent ID, systemd unit, host group, log selector, metrics namespace, trace service name, preferred source, analysis profile.
- Restart policy: allow_restart, requires_human_approval, cooldown minutes, allowed service types.

Certification stages: catalog_only, observe_only, correlate_ready, diagnostics_ready, restart_ready. Do not skip stages.

What to avoid

Do not add vague services like "IDC stuff" or "Core systems." Vague objects create vague incidents, false correlation, and unsafe recommendations.

Operator trick

When Nexus feels wrong, compare service_id, Network Sentinel mapping, agent ID, cluster membership, flow steps, and dependency edges before changing incident logic.

Databases are operational citizens, not notes on an app

A database can be the root cause, a shared dependency, a blast-radius multiplier, or a restart blocker. Model it explicitly so Nexus can distinguish app failure from data-store pressure.

DATABASE CONTRACT

Fields Nexus expects

- enabled, platform, host, database_name, instance_name, service_name, username, JDBC URL, connection type.
- role, host group, port, schemas, connection pool, max pool size.
- replication group, failover group, read-only state, shared dependency, data classification.
- expected evidence and safe diagnostics.
- metadata for local realities that do not fit normal fields.

If a database is shared or operationally important, add it as its own db service and connect dependent apps to it with db edges.

DATABASE PROFILE

Oracle PROD

Evidence
sessions

Evidence
locks

Evidence
ORA/TNS

Schemas

IDC, AUDIT

Pool

Hikari, max 60

Safe probes

connectivity, sessions, locks, lag

Do not

restart from Nexus

DB edge details

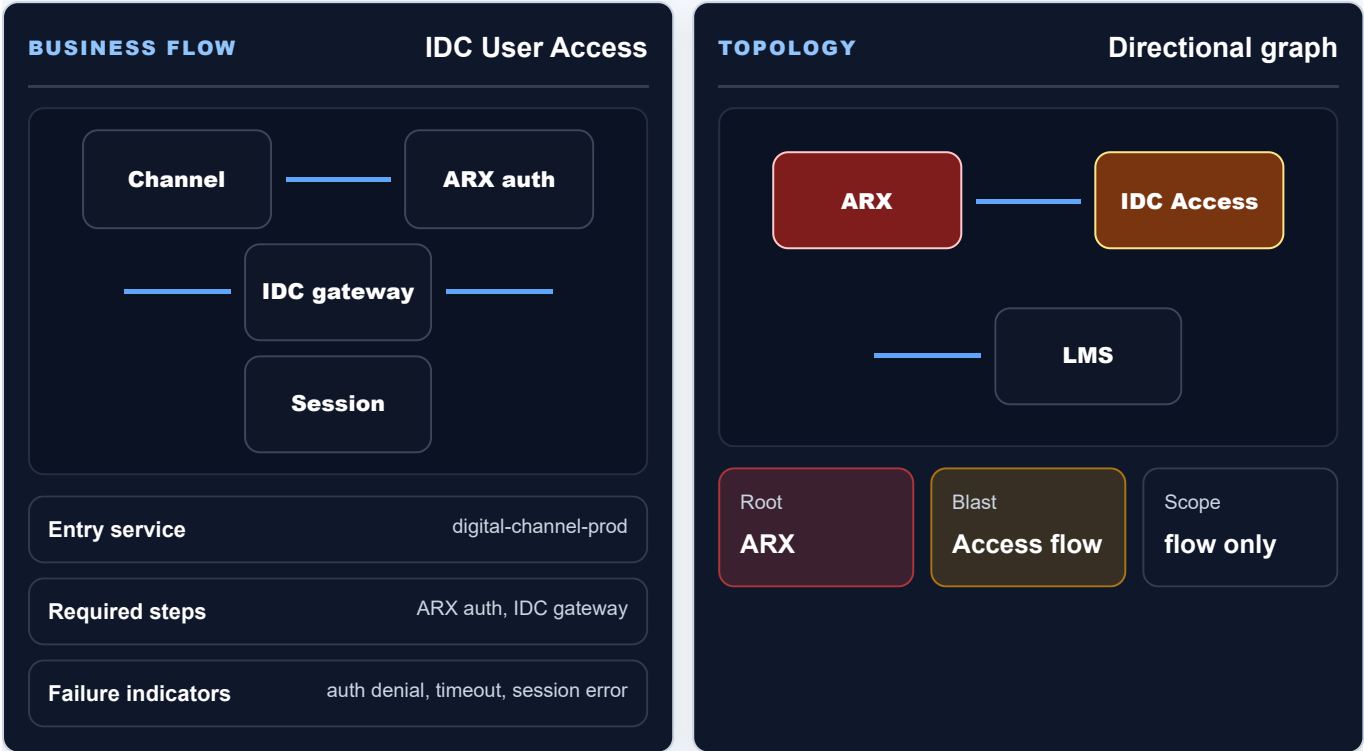
- access_mode, schema_names, operation_types.
- connection_pool, max_connections, statement_timeout_ms.
- expected_error_codes: SQLSTATE, ORA, TNS, JDBC.
- query_fingerprint_scope and transactional state.

How to read DB evidence

- High sessions plus pool saturation means pressure may be below the app.
- Locks and deadlocks point to contention, not necessarily service death.
- Replication lag can cause stale reads or failover symptoms.
- Tablespace or storage pressure can look like random application failure.

Model real journeys and directional dependencies

Clusters group operational fabrics. Business flows describe user and transaction journeys. Dependency edges explain direction, purpose, scope, failure domain, expected evidence, and timeout budget.



Cluster

Use for operational fabrics: IDC core, LMS processing, ARX access, Postilion card flow, digital channels, infrastructure.

- Entry services
- Service IDs
- Routing URLs
- Owner and criticality

Business flow

Use for real journeys. A service can appear in multiple flows with different meaning.

- Flow ID and name
- Ordered steps
- Required roles
- Success and failure indicators

Dependency edge

Use for directional truth. Edges drive root cause, blast radius, and recommendations.

- From and to service
- Type and purpose
- Global or flow-scoped
- Expected evidence

Trick: Use flow_scoped edges when the dependency is true only inside a specific journey. Use global only when the dependency truly applies everywhere.

Runtime evidence and control must stay bounded

Light agents are the bridge between Nexus and host-local reality. They report state, logs, diagnostics readiness, service mapping, and command capability. They must never become arbitrary remote shell.

AGENTS WORKSPACE **agent-prod-02**

Heartbeat
14s ago

Configured
7 services

Reporting
6 services

Capabilities logs, diagnostics, metrics

Missing idc-worker-prod

Unexpected old-api-prod

Command surface allowlisted only

AGENT CHECKLIST

Before trusting runtime evidence

- Confirm token status, heartbeat age, host, resource, command server, and service list.
- Compare configured service count with reporting service count.
- Fix missing service IDs and unexpected service IDs before using evidence for decisions.
- Confirm command output is bounded and service-specific.
- Never accept arbitrary commands from operators, incident notes, log lines, prompts, or request payloads.

Agent presence is not restart permission. Restart readiness is separate certification plus policy gate.

Service live state

- Status, agent, network, signals.
- Dependencies and diagnostics state.
- Recent actions and control readiness.
- Source explorer and log tail for focused evidence.

What to avoid

- Do not attach one generic agent identity to multiple unrelated services without mapping.
- Do not use manual local server knowledge as a substitute for Nexus service contracts.
- Do not run undocumented commands outside the guarded control lane.

Govern the procedures that Copilot can cite

Nexus procedure guidance works best when SOPs are accurate, scoped, validated, approved, and attached to real services and failure modes.

TWO SOP PLANES

Indexed corpus and managed registry

- **Indexed Copilot Corpus:** procedures retrieval can cite today, including legacy/manual SOPs.
- **Managed Registry:** database-owned procedures admins can edit, validate, approve, deprecate, and re-index.
- If Copilot cites an indexed-only SOP, adopt it into the managed registry before editing and governing it.

Ask Copilot from a real incident so it can use affected services, root candidates, evidence timeline, business flow, failure domain, and recommendations.

SOP WORKSPACE

Governed procedure

Indexed corpus	Managed registry
Copilot can cite	Admin can approve
Scope	IDC gateway, production
Preconditions	incident open, evidence confirmed
Unsafe actions	no restart for DB pressure
Validation	auth test, gateway health, verdict

SOP approval checklist

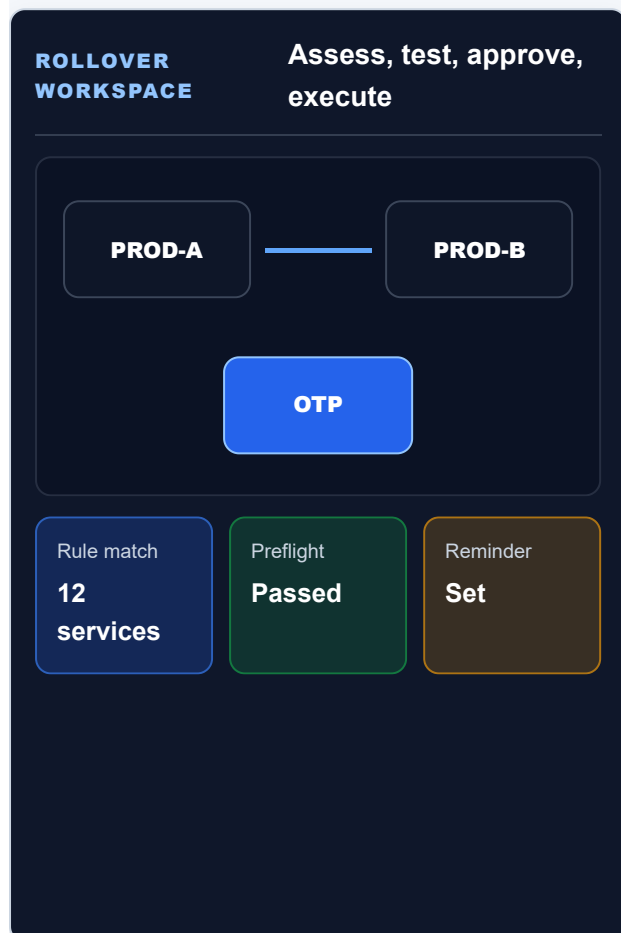
- Owner, service scope, environment, and affected business flows.
- Preconditions, checks, action steps, rollback, escalation, and validation.
- Unsafe-action warnings and when to stop.
- Retrieval citations and confidence tested from a real incident.

Operator trick

A SOP that tells operators what to do but not when to avoid it is incomplete. Add unsafe-action warnings before approval.

Environment rollover is a governed change, not a hidden edit

The rollover workspace controls environment profile movement, replacement rules, assessments, execution, OTP approval, and reminders. Use it when services, connections, or environments are being replaced in a controlled way.



ROLLOVER CHECKLIST

How to move safely

1. Confirm source and target environment profiles.
2. Review replacement rules and assessment output.
3. Inspect impacted services, endpoints, connection profiles, and agent mapping.
4. Run preflight tests and record approval reason.
5. Complete OTP challenge and execute only after evidence is ready.
6. Monitor execution result and set reminders for follow-up verification.

Do not use rollover as a shortcut for undocumented config changes. It must leave an operational record.

Common failure after rollover

Old connection profiles remain in an agent, service endpoint, or dependency edge. Nexus then sees false runtime or dependency failures.

Correct operator move

Run assessment again, compare old and new profile references, verify light-agent service config, and record the final state in the incident or change notes.

How experienced operators read Nexus

The same alert can mean different things depending on service contract, topology, evidence source, flow scope, and database context. These examples show the reading pattern.

IDC access issue with ARX

Scenario: IDC login failures appear, ARX auth errors spike, IDC runtime is healthy.

Read: ARX is likely for access if the edge is flow-scoped to IDC User Access.

Move: Diagnose ARX, create task for auth owner, do not blame ARX for unrelated transaction posting.

DB session exhaustion

Scenario: App timeouts rise, external checks are green, database sessions and pool usage are critical.

Read: App may be symptomatic; database pressure is probable root.

Move: Request DB-safe diagnostics and alert database owner instead of restarting apps repeatedly.

Network path outage

Scenario: External timeout appears but local agent reports healthy process and clean logs.

Read: Reachability may be failing outside runtime: VIP, DNS, route, firewall, load balancer, or monitoring path.

Move: Create a network response task and avoid restart until runtime evidence supports it.

USSD session expiry burst

Scenario: Compact multi-carrier session expiry bursts appear on txn-mobile-ussd.

Read: With the correct service-specific profile, this can be channel_tunnel evidence. Isolated expiries can be normal inactivity.

Move: Verify the profile is attached only to USSD and correlate with carrier and gateway evidence.

Postilion card chain

Scenario: Direct Channel, Postilion, and IDC all show card transaction symptoms.

Read: Earliest compatible signature and graph direction decide root versus downstream symptom.

Move: Read topology before assigning owner. Create task for the earliest supported failure point.

Rollover drift

Scenario: PROD-A was replaced by PROD-B, but an agent or service still references old profiles.

Read: False dependency failures may appear after an incomplete rollover.

Move: Re-run assessment, verify services and agents, then record final state.

The Nexus discipline

The best Nexus operator does not click fastest. The best Nexus operator keeps truth aligned: catalog truth, graph truth, evidence truth, procedure truth, action safety, and outcome truth.

Before investigation

- Read fabric pulse and sync health.
- Check mapped service coverage.
- Confirm the incident has correct services, flows, and clusters.
- Open the highest-risk business impact first.

Before action

- Read topology direction and flow scope.
- Compare external_network, runtime, log, trace, and DB evidence.
- Use diagnostics when the story is plausible but incomplete.
- Create tasks for owned follow-up.

Before restart

- Target is exact, stateless, restart_ready, and policy allowed.
- No maintenance or cooldown blocks exist.
- Evidence supports restart as safer than waiting or escalation.
- Human approval is complete.

Before closing

- Record actual root service.
- Record action taken and whether it worked.
- Mark false positive, duplicate, early warning, confirmed, or wrong root cause.
- Write notes that the next operator can understand.

GOLDEN RULE

A smaller verified topology is better than a large confident lie

Build Nexus one verified service, one verified flow, and one verified dependency at a time. Every correct record makes Nexus sharper. Every vague record makes it louder.

DOWNLOAD AND USE

Keep this guide next to the page

Use the in-app Nexus Guide for quick help while operating. Use this PDF for onboarding new operators, training admins, and reviewing the complete Nexus operating model.